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Motion

Motion

If we look around us, we find that there are number of objects which are in motion.

An object is said to be in motion if it changes its position with the passage of time.

Now observe the following bodies or objects to understand the meaning of the term "motion". Cars, cycles, motorcycles, scooters, buses, rickshaws, trucks, etc. running on the road, birds flying in the sky, fish swimming in water, all these objects are in motion. Very small objects like atoms, molecules and very large objects like planets, stars and galaxies are also in motion.

Thus, all objects ranging from the smallest atom to the largest galaxy are in continuous motion.

Kinematics is the science of describing the motion of objects using words, diagrams, numbers, graphs, and equations.

"Motion is the change in position of an object with time."

Concept of a point object (or particle)

► Point object

An extended object can be treated as a point object when the distance travelled by the object is much greater than its own size.

A point object (or particle) is one, which has no linear dimensions but possesses mass.

Examples : (i) Study of motion of a train travelling from Kota to New Delhi.

(ii) Revolution of earth around the sun for one complete revolution.

Concept of an extended object

► Extended object

An object whose dimensions can be measured. **Example :** Car, bus, train, etc.

Describing motion

When a tree is observed by an observer A standing at the railway station, the tree is at rest. This is because the position of the tree is not changing with respect to the observer A (see figure).



Train moving on a railway track

Now, when the same tree is observed by an observer B sitting in a superfast train moving with a velocity v , then the tree is moving with respect to the observer because the position of the tree is changing with respect to the observer B.

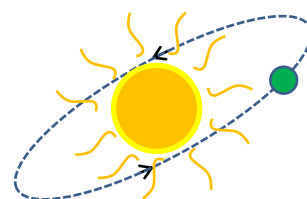
Rest and motion are relative terms : There is nothing like absolute rest or motion. This means that an object can be at rest and also in motion at the same time i.e. all objects, which are stationary on earth, are said to be at rest with respect to each other, but with respect to the sun they are making revolutions. In order to study motion, therefore, we have to choose a fixed position or point with respect to which the motion has to be studied. Such a point or fixed position is called a **reference point** or the **origin**.



Discuss whether the walls of your classroom are at rest or in motion.

Explanation

The walls of our classroom are at rest with respect to the ground or earth. But, they are in motion with respect to an object or an observer outside the earth. This is because the earth is moving about its own axis as well as it is revolving around the sun. Thus, the state of rest and motion are not absolute, they are relative terms.



Example

When you sit on a chair, your speed is zero relative to Earth but 30 km/s relative to the Sun.

Types of motion

Translational motion (or translatory motion)

Motion of a body in which all the points in the body follow parallel paths is called 'translational motion'. It is a motion in which the orientation of an object remains the same throughout the journey. The path of a translatory motion can be straight or curved.

Examples of translational motion

- | | |
|--------------------------------------|---|
| (1) A car moving down a highway. | (2) A person walking on the road. |
| (3) An athlete running on the track. | (4) Motion of a piston inside the cylinder. |
| (5) A train running on the rails. | |

On the basis of the path travelled by an object, the translational motion can be classified as :

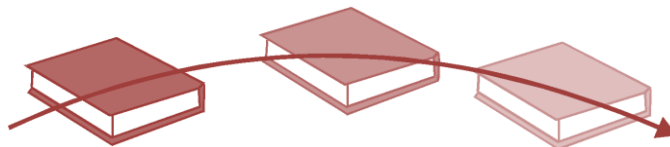
- (1) Rectilinear motion :** If an object moves in a straight line, its motion is called rectilinear motion or one dimensional motion (see figure). Motion of a car along a straight path, motion of a piston in the cylinder are examples of rectilinear motion.



A book moved along a straight path without changing its orientation

Rectilinear motion

- (2) **Curvilinear motion** : If an object moves along a curved path without change in its orientation, its motion is called curvilinear motion (see fig). Motion of a car along a curved or circular path, motion of an athlete on a circular track are examples of curvilinear motion.

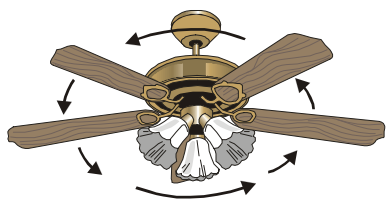


A book moved along a curved path without changing its orientation

Curvilinear motion

Rotational motion (Rotatory motion)

Motion of a body turning about an axis is called rotational motion. In other words, 'a motion in which an object spins about a fixed axis is called rotational motion'. It is a motion in which the orientation of an object continuously changes throughout the motion. The path of an object in a rotational motion is always circular (or curved).



(a) Motion of a ceiling fan



(b) Motion of Earth about its axis



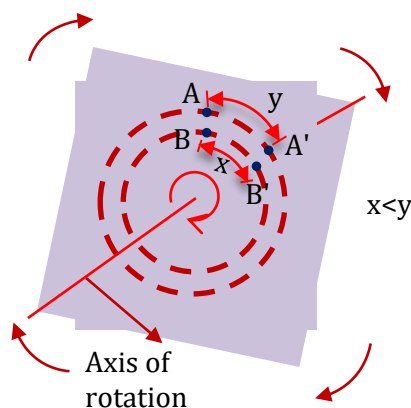
(c) Motion of a spinning top

Examples of rotational motion

Examples of rotational motion

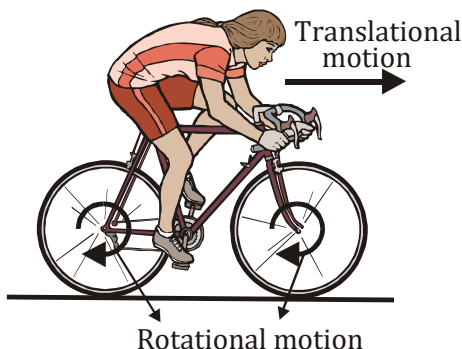
- (1) The Earth's spin on its axis.
- (2) Motion of a fan or motor.
- (3) Motion of the blades of windmill.
- (4) Motion of a spinning top.
- (5) Motion of a grinding stone.

- In rotational motion, the particles of the object move through the unequal distances in a given time depending on their location in the object (see figure).



Rotational motion : Particles cover unequal distances in a given time.

The particle which is located near the axis of rotation, covers less distance as compared to the particle that is located far away from the axis. Motion of a car or bicycle wheels is a combination of translational and rotational motion (see figure).



Motion of a bicycle wheel is a combination of translational and rotational motion, also called 'rolling motion'.

Circular motion

When an object moves along a circular path, this type of curvilinear motion is called 'circular motion'. For example, motion of an athlete along a circular track is a circular motion.

Periodic motion

A motion that occurs when an object moves in a repeated pattern (a cycle) over equal periods of time is called a **periodic motion**.

Examples : Motion of a pendulum, rotational motion of Earth, revolution of Earth around the Sun are all periodic motions.

Oscillatory motion

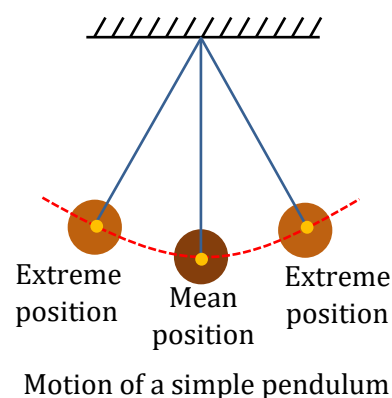
A motion that occurs when an object moves to and fro about its mean position over equal periods of time is called an **oscillatory motion** or **vibratory motion**.

Examples : Motion of a simple pendulum, motion of a vibrating stretched string, motion of an oscillating spring.

Mean position

A position in the path of an oscillating particle at which the net force acting on the particle is zero is called its equilibrium position or mean position.

Examples : The vertical position or the lowest position of an oscillating pendulum is its equilibrium position or mean position (see figure). When a pendulum is moved away from the mean position and then released, then it again come towards the mean position and starts oscillating about its mean position. It moves and reaches extreme positions on both sides of the mean position.



Scalar and vector quantities

Scalar quantity

A physical quantity that is defined by its magnitude only is called a scalar quantity.

Examples : Mass, time, distance, speed, work, power, energy, electric charge, volume, density, pressure, electric potential, temperature, etc.

Scalar quantities follow the algebraic (scalar) laws of addition.

Vector quantity

A physical quantity that is defined by its magnitude as well as direction and follows the vector laws of addition is called a vector quantity.

Examples : Velocity, acceleration, force, displacement, momentum, weight, torque, electric field, magnetic field, etc.

Arrows (or rays) are used to represent vectors. The direction of the arrow gives the direction of the vector. The length of the arrow is proportional to the magnitude of the vector.



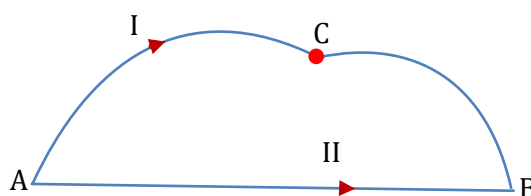
Two vectors are equal if they have the same magnitude and the same direction.

Distance and displacement

Distance

The length of the actual path between the initial and the final position of a moving object in the given time interval is known as the distance travelled by the object.

Distance = Length of path I (ACB) (see figure)



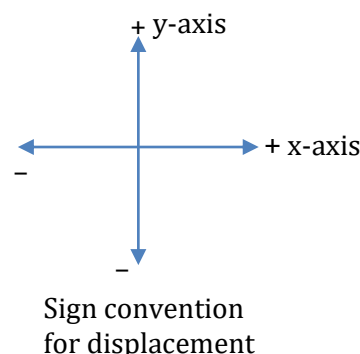
Concept of distance and displacement

- ▶ Distance is a **scalar** quantity. It is always taken positive.
- ▶ Distance is measured by an odometer in vehicles.
- ▶ **Units : In SI system :** metre (m)

In CGS system : centimetre (cm)

Displacement

- ▶ The shortest distance between the initial position and the final position of a moving object in the given interval of time is known as the displacement of the object.
- ▶ Displacement = Length of path II (AB) (see figure)
- ▶ Displacement of an object may also be defined as the change in position of the object in a particular direction. That is,



► **Displacement of an object = Final position – Initial position of the object = $x_f - x_i$**

► Displacement is a **vector** quantity. Displacement can be positive, negative or zero.

► **Units : In SI system : metre (m)**

In CGS system : centimetre (cm)

Important points related to distance and displacement

► During motion, displacement of an object may be zero but the distance travelled by the object is never zero.

► Distance travelled by an object is either equal to or greater than the magnitude of displacement of the object.

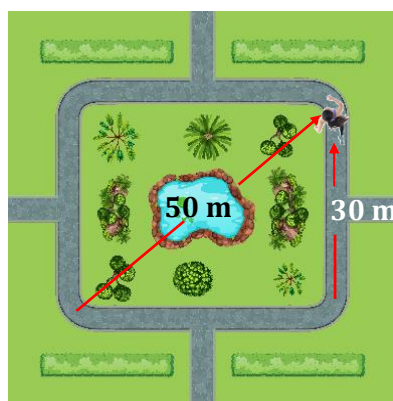
► Distance is equal to the magnitude of displacement when a body moves in a straight line in a particular direction, or it is in uniform motion.

Example

Suppose the distance you traveled from your house to the park was 200 m. When you get to the park, how would you describe your location? You could say that your location was 200 m from your house. However, your final position depends on both the distance you travel and the direction. Did you go 200 m east or west? To describe your final position exactly, you also would have to tell the direction from your starting point. To do this, you would specify your displacement. Displacement includes the distance between the starting and the ending points and the direction in which you travel. Figure shows the difference between distance and displacement.



Distance 40 m
Displacement : 40 m east



Distance 70 m
Displacement : 50 m north east



Distance 140 m
Displacement : 0 m



A honeybee leaves the hive and travels 2 meters as it returns to the hive. Is the displacement for the trip the same as the distance travelled? If not, why not?



Explanation

No, the displacement and the distance are not same. This is because the displacement is the change in position of object in motion while distance is the length of path travelled by it.

Here, the distance travelled = 2 m

While, the displacement = 0, because the position of honey bee has not changed.



- ★ **For a Circle :** Diameter = $2r$

$$\text{Circumference} = 2\pi r \text{ or } \pi d \left(\pi = \frac{22}{7} \text{ or } 3.14 \right)$$

- ★ **Quadrilateral**

$$\text{Perimeter of quadrilateral} = AB + BC + CD + DA$$

$$\text{Perimeter of rectangle} = 2(l + b)$$

$$\text{Perimeter of square} = 4 \times \text{side}$$

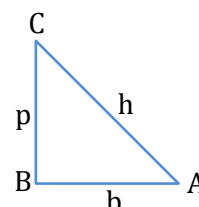
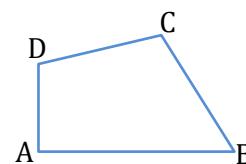
- ★ **Pythagoras theorem**

$$(\text{Hypotenuse})^2 = (\text{Perpendicular})^2 + (\text{Base})^2$$

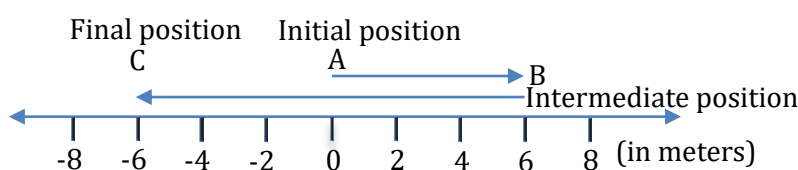
$$AC^2 = BC^2 + AB^2$$

- ★ A body moving in a circular path when covers $\frac{1}{4}$ th of the circle then, the distance travelled by it is $\frac{\pi r}{2}$ and displacement is $r\sqrt{2}$.

- ★ A body moving in a circular path when covers $\frac{3}{4}$ th of the circle then, the distance travelled by it is $\frac{3\pi r}{2}$ and displacement is $r\sqrt{2}$.



1. **Motion of a particle is shown below on a number line. Find the displacement from (a) A to B (b) B to C (c) overall journey. Also, find distance for overall journey.**



Solution

(a) Displacement from A to B = $x_f - x_i = (6) - (0) = +6 \text{ m}$

(b) Displacement from B to C = $x_f - x_i = (-6) - (6) = -12 \text{ m}$

(c) Displacement of overall journey (i.e. A to B, B to C) = $x_f - x_i = (-6) - (0) = -6 \text{ m}$

Here, distance can also be found by adding positive values of displacement AB & displacement BC. i.e.,

$$\text{Distance travelled during overall journey} = AB + BC = 6 + 12 = 18 \text{ m}$$

Here Distance > |Displacement|

Whenever a particle changes its direction, distance is greater than the magnitude of displacement.

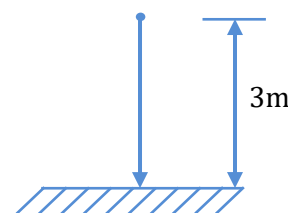
2. A body falls from a height of 3 m.

Find displacement and distance.

Solution

Displacement = $x_f - x_i = 0 - 3 = -3\text{ m}$, Distance = **3 m**

(see figure)



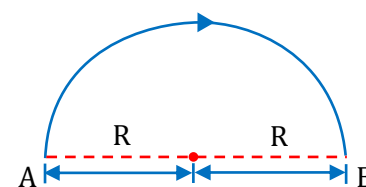
3. A particle moves along a circular path as shown in figure.

Find distance travelled and displacement.

Solution

$$\begin{aligned} \text{Distance travelled} &= \frac{1}{2} \times (\text{circumference of the circle}) \\ &= \frac{1}{2} (2\pi R) = \pi R, \text{ and} \end{aligned}$$

Displacement = diameter AB = **2R**



- A body moving in a circular path when reaches its original position after one round, then the displacement at the end of one round is zero, but the distance travelled by it is equal to the circumference of circular path.

4. In a long-distance race, the athletes were expected to take four rounds of the track such that the line of finish was same as the line of start. Suppose the length of the track was 200 m.

(i) What is the total distance to be covered by the athletes?

(ii) What is the displacement of the athletes when they touch the finish line?

(iii) Is the displacement of an athlete and the distance moved by him at the end of the race equal?

Solution

(i) Total distance covered = $4 \times 200 \text{ m} = \mathbf{800 \text{ m}}$

(ii) As the athletes finish at the starting line,

Displacement = final position - initial position = $x_A - x_A = \mathbf{0}$

(iii) Displacement and distance moved are not equal.

5. Find the distance and displacement travelled by the object shown in figure while moving from A to B.

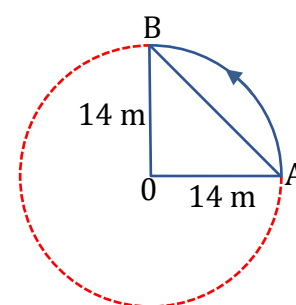
Solution

$$\text{Distance} = \frac{1}{4} \times 2\pi r = \frac{1}{4} \times 2 \times \frac{22}{7} \times 14 = \mathbf{22 \text{ m}}$$

Displacement = length of line AB

Since, $\triangle AOB$ is a right angled triangle.

$$AB^2 = OA^2 + OB^2 \Rightarrow AB^2 = (14)^2 + (14)^2 \Rightarrow AB = \sqrt{2 \times (14)^2} = \mathbf{14\sqrt{2} \text{ m}}$$



6. Find the distance and displacement travelled by the object shown in figure while moving from A to B.

Solution

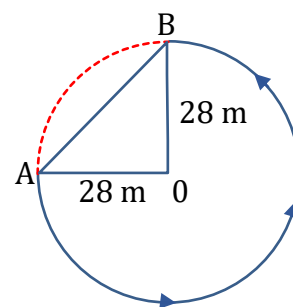
$$\begin{aligned}\text{Distance} &= \frac{3}{4} \times 2\pi r \\ &= \frac{3}{4} \times 2 \times \frac{22}{7} \times 28 = 132 \text{ m}\end{aligned}$$

Displacement = length of AB,

Since, $\triangle AOB$ is a right angled triangle

$$AB^2 = OA^2 + OB^2 \Rightarrow AB^2 = (28)^2 + (28)^2$$

$$\Rightarrow AB = \sqrt{2 \times (28)^2} = 28\sqrt{2} \text{ m}$$



Speed

Speed of a body is the distance travelled by the body per unit time. The rate of change of distance is called speed.

$\text{speed} = \frac{\text{distance travelled}}{\text{time taken}}$
--

If a body covers a distance s in time t then speed, $v = \frac{s}{t}$

Unit

In SI system : m/s or ms^{-1}

In CGS system : cm/s or cms^{-1}

A commonly used unit of speed is km/h or kmh^{-1} .

- Speed is a **scalar** quantity, because it has magnitude but no direction. Speed is always taken non negative (positive or zero)

Uniform speed

When a body covers equal distances in equal intervals of time, the body is said to be moving with a uniform speed or constant speed.

- Examples :**
- (i) A train running with a speed of 120 km/h.
 - (ii) An aeroplane flying with a speed of 600 km/h.

Non-uniform speed

When a body covers unequal distances in equal intervals of time, the body is said to be moving with non-uniform speed or variable speed.

- Examples :**
- (i) A car running on a busy road.
 - (ii) An aeroplane landing on a runway.

Average speed

The average speed of the body in a given time interval is defined as the total distance travelled divided by total time taken.

$$\text{Average speed} = \frac{\text{Total distance travelled}}{\text{Total time taken}}$$

$$\frac{1 \text{ km}}{\text{h}} = \frac{1000 \text{ m}}{60 \times 60 \text{ s}} = \frac{5}{18} \text{ m/s}$$

$$\text{Speed in m/s} = \frac{5}{18} \times \text{speed in km/h ;}$$

$$\text{km h}^{-1} \xrightarrow[\times \frac{5}{18}]{\times \frac{18}{5}} \text{ms}^{-1}$$



1. A car travels first half distance with a uniform speed u and next half distance with a uniform speed v . Find its average speed.

Solution

$$\text{Total distance} = \frac{d}{2} + \frac{d}{2} = d \quad [\text{See figure}]$$

$$\text{Total time} = t_1 + t_2 = t$$

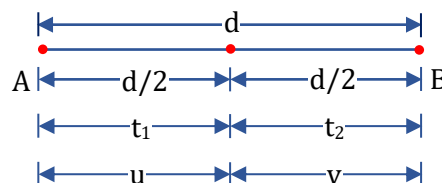
$$\therefore t_1 = \frac{d/2}{u} \quad \dots(i) \quad \left[t = \frac{s}{v} \right]$$

$$t_2 = \frac{d/2}{v} \quad \dots(ii)$$

$$V_{av} = \frac{d}{t} \quad \text{Putting the value of equation (i) and (ii),}$$

$$V_{av} = \frac{d}{\frac{d/2}{u} + \frac{d/2}{v}} = \frac{d}{\frac{d}{2} \left(\frac{1}{u} + \frac{1}{v} \right)} = \frac{2}{\frac{1}{u} + \frac{1}{v}}$$

$$\therefore V_{av} = \frac{2uv}{u+v}$$



2. A car travels first half time with a uniform speed u and next half time with a uniform speed v . Find its average speed.

Solution

$$d_1 = u \times \frac{t}{2}, \quad d_2 = v \times \frac{t}{2} \quad [\text{See figure}]$$

$$\text{Total distance } d = d_1 + d_2 = u \times \frac{t}{2} + v \times \frac{t}{2}$$

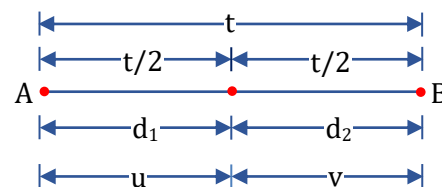
$$d = \frac{ut}{2} + \frac{vt}{2} = \frac{t}{2}(u+v)$$

$$\text{Total time} = t ;$$

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time taken}}$$

$$V_{av} = \frac{\frac{t}{2}(u+v)}{t}$$

$$\therefore V_{av} = \frac{u+v}{2}$$



Instantaneous speed

The speed of a body at any particular instant of time during its motion is called the instantaneous speed of the body. It is measured by **speedometer** in vehicles.



1. On a 120 km track, a train travels the first 30 km with a uniform speed of 30 km/h. How fast must the train travel the next 90 km so as to average 60 km/h for the entire trip?

Solution

Given; Total distance $d = 120$ km, Average speed $V_{av} = 60$ km/h, Total time $= t = ?$

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time taken}} = V_{av} = \frac{d}{t} \quad \text{or} \quad t = \frac{d}{V_{av}}$$

Putting the values,

$$t = \frac{120 \text{ km}}{60 \text{ km/h}} = 2 \text{ h} \quad \dots(i)$$

Distance travelled in first part of trip, $d_1 = 30$ km, Speed in first part of the trip, $v_1 = 30$ km/h

$$\text{Time taken in first part of trip, } t_1 = \frac{d_1}{v_1}$$

$$\text{Putting the values, } t_1 = \frac{30 \text{ km}}{30 \text{ km/h}} = 1 \text{ h}$$

Time taken to complete second part of the trip, $t_2 = t - t_1 = 2 - 1 = 1$ h

Distance to be covered in second part of the trip, $d_2 = 90$ km,

Required speed in second part, $v_2 = ?$

$$\text{Speed} = \frac{\text{distance}}{\text{time}}, \quad \therefore v_2 = \frac{d_2}{t_2} = \frac{90 \text{ km}}{1 \text{ h}} = 90 \text{ km/h}$$

2. A bus going from Kota to Jaipur passed the 100 km, 160 km and 220 km points at 10:30 am, 11:30 am and 1:30 pm. Find the average speed of the bus during each of the following intervals:

(a) 10.30 am to 11.30 am, (b) 11.30 am to 1.30 pm and (c) 10.30 am to 1.30 pm.

Solution

(a) The distance covered between 10:30 am and 11:30 am is $160 \text{ km} - 100 \text{ km} = 60 \text{ km}$.

The time interval is 1 hour. The average speed during this interval is –

$$v_1 = \frac{60 \text{ km}}{1 \text{ h}} = 60 \text{ km/h}$$

(b) The distance covered between 11:30 am and 1:30 pm is $220 \text{ km} - 160 \text{ km} = 60 \text{ km}$. The time interval is 2 hours. The average speed during this interval is –

$$v_2 = \frac{60 \text{ km}}{2 \text{ h}} = 30 \text{ km/h}$$

(c) The distance covered between 10:30 am and 1:30 pm is $220 \text{ km} - 100 \text{ km} = 120 \text{ km}$.

The time interval is 3 hours. The average speed during this interval is –

$$v_3 = \frac{120 \text{ km}}{3 \text{ h}} = 40 \text{ km/h}$$

3. The average speed of a bicycle, an athlete and a car are 18 km/h, 7 m/s and 2 km/min. respectively. Which of the three is the fastest and which is the slowest?

Solution

$$18 \text{ km/h} = \frac{18 \text{ km}}{1 \text{ h}} = \frac{18000 \text{ m}}{3600 \text{ s}} = 5 \text{ m/s}$$

$$2 \text{ km/min} = \frac{2 \text{ km}}{1 \text{ min.}} = \frac{2000 \text{ m}}{60 \text{ s}} = 33.3 \text{ m/s}$$

Thus, the average speeds of the bicycle, the athlete and the car are 5 m/s, 7 m/s and 33.3 m/s respectively. So the car is the fastest, and the bicycle is the slowest.

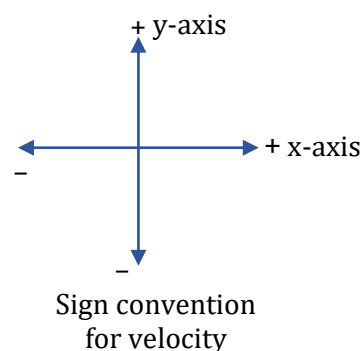
Velocity

The velocity of a body is the displacement of a body per unit time.

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}}$$

The displacement covered by a body per unit time or the speed of a body in a specified direction is called **velocity**.

► Velocity is a **vector** quantity. It can be positive, negative or zero (see figure).



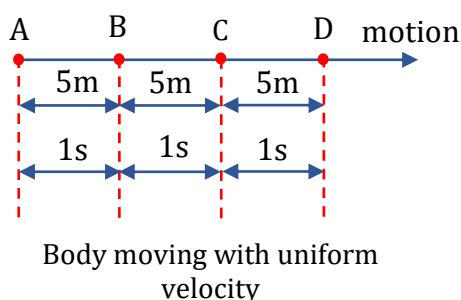
Unit

In SI system : m/s or ms⁻¹

In CGS system : cm/s or cms⁻¹

Uniform velocity

When a body covers equal displacements in equal intervals of time in a particular direction, the body is said to be moving with a uniform velocity (see figure).



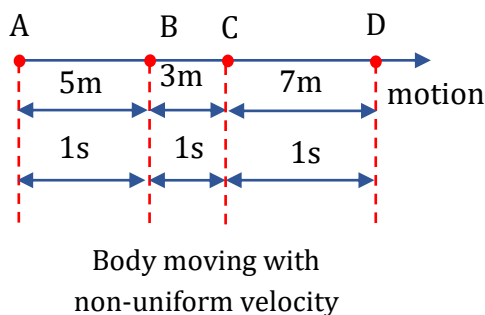
Conditions for uniform velocity

- The body must cover equal displacements in equal intervals of time.
- The direction of motion of the body should not change.

Example : A train running towards south with a speed of 120 km/h.

Non-uniform velocity/variable velocity

When a body covers unequal displacements in equal intervals of time, the body is said to be moving with variable velocity (see figure).



When a body covers equal distances in equal intervals of time, but its direction changes, then the body is said to be moving with variable velocity.

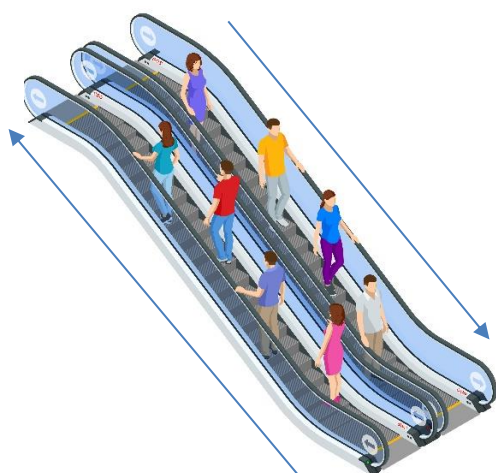
Example : In circular motion, a particle may have constant speed but its direction changes continuously thus, its velocity is non-uniform.

Conditions for variable velocity

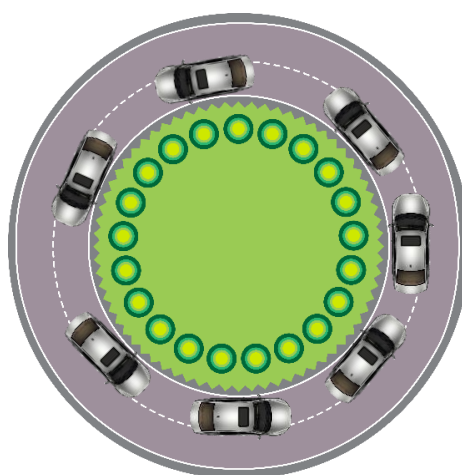
- (i) It should cover unequal displacements in equal intervals of time.
- (ii) It should cover equal distances in equal intervals of time but its direction must change.

Examples

- (i) A car running towards north on a busy road has a variable velocity as the displacement covered by it per unit time changes with change in the road condition.
- (ii) The blades of a rotating ceiling fan, a person running around a circular track with constant speed etc. are the examples of variable velocity.



The people on these two escalators have the same speed. However, their velocities are different because they are travelling in opposite directions.



The velocity of this car changes because its direction of motion is changing.

Average velocity

Total displacement of a particle divided by total time taken is called average velocity.

$$\text{Average velocity} = \frac{\text{Total displacement}}{\text{Total time taken}}$$

$$V_{av} = \frac{s_2 - s_1}{t_2 - t_1}$$

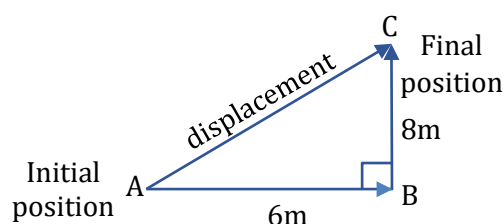
Average speed is always greater than or equal to magnitude of average velocity. Average speed is equal to average velocity when particle moves in a straight line without change in direction.

Instantaneous velocity

The velocity of a body at any particular instant of time during its motion is called the instantaneous velocity of the body.



A particle moves along a path ABC as shown in figure. The time taken during the journey is 2 seconds. Find the average speed and average velocity during the journey.



Solution

Total distance travelled, $s = AB + BC = 6 + 8 = 14$ m

$$\text{Average speed, } V_{av} = \frac{s}{t} = \frac{14}{2} = 7 \text{ m/s}$$

Here, displacement, $\vec{s} = AC = \sqrt{AB^2 + BC^2} = \sqrt{(6)^2 + (8)^2} = \sqrt{100} = 10$ m

$$\therefore \text{Average velocity, } \vec{V}_{av} = \frac{\vec{s}}{t} = \frac{10}{2} = 5 \text{ m/s}$$

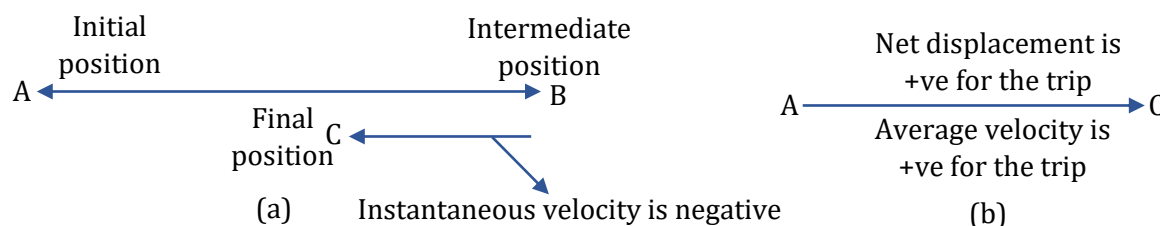
- Here, average speed is greater than average velocity because the direction of particle changes during motion.



1. The average velocity for a trip has a positive value. Is it possible for the instantaneous velocity at any point during the trip to have a negative value? Justify your answer.

Explanation

Yes, the average velocity for a trip has a positive value but its instantaneous velocity at any point may be negative. This is possible when the particle or object has reversed its direction and moved towards the initial position but stopped before reaching the initial position (see figure).



Uniform and non-uniform motion

A moving body may cover equal distances in equal intervals of time or different distances in equal intervals of time. On the basis of above assumption, the motion of a body can be classified as uniform motion and non-uniform motion.

Uniform motion

When a body covers equal distances in equal intervals of time, however small may be the time intervals, in a particular direction, the body is said to describe **a uniform motion**.

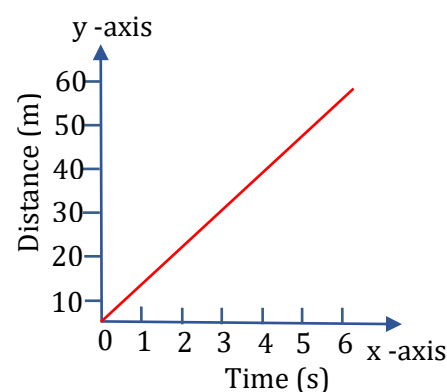
Time (in second)	0	1	2	3	4	5	6
Distance covered (in metre)	0	10	20	30	40	50	60

- Uniform motion always takes place in a straight line.

In uniform motion, velocity of particle remains constant i.e., its magnitude as well as direction are constant. In uniform motion, average speed/velocity is equal to instantaneous speed/velocity at any point of time.

Examples of uniform motion

- An aeroplane flying at a speed of 600 km/h along north.
- A train running at a speed of 120 km/h along east.
- Light energy travelling at a speed of 3×10^8 m/s in vacuum.



Distance-time graph for uniform motion

Non-uniform motion

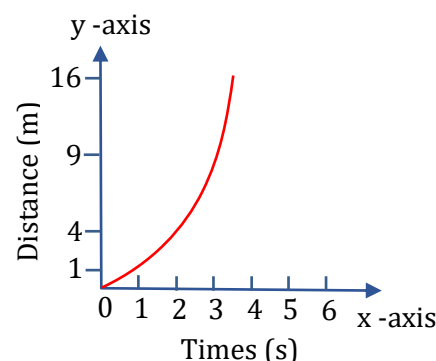
When a body covers unequal distances in equal intervals of time, the body is said to be moving with **a non-uniform motion**.

- Any motion along a curved path is always non-uniform motion. Also, any motion in which particle changes its direction is also non-uniform motion.

Time (in second)	0	1	2	3	4
Distance (in metre)	0	1	4	9	16

Examples of non-uniform motion

- An aeroplane running on a runway before taking off.
- A freely falling stone under the action of gravity.
- When the brakes are applied to a moving car.
- A fan rotating with constant speed is also in non-uniform motion.



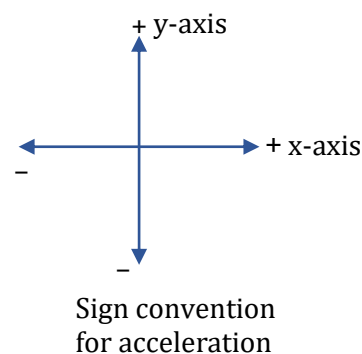
Distance-time graph for non-uniform

Acceleration

In uniform motion, the velocity remains constant with time. Thus, the change in velocity for any time interval is zero. But, in non-uniform motion, velocity changes with time. Thus, the change in velocity for any time interval has a non-zero value.

In non-uniform motion, a new physical quantity called 'acceleration' is introduced.

The rate of change of velocity of a moving body with time is called **acceleration**.



$$\text{Acceleration} = \frac{\text{Change in velocity}}{\text{Time taken for change}}$$

But, change in velocity = final velocity – initial velocity.

$$\text{Acceleration} = \frac{\text{Final velocity} - \text{Initial velocity}}{\text{Time taken for change}} = a = \frac{v - u}{t}$$

If body moves with uniform velocity, then $v = u$ and acceleration is zero i.e. $a = 0$.

► Acceleration is a vector quantity. It can be negative, positive or zero.

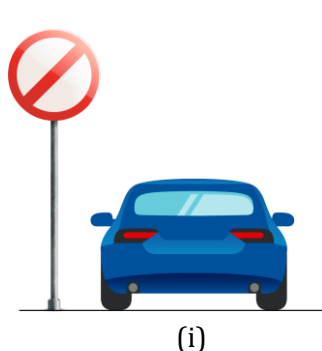
Unit of acceleration

In SI system : m/s^2 or ms^{-2}

In CGS system : cm/s^2 or cms^{-2}

Examples of acceleration :

- A car speeding up as it leaves the town. The driver presses on the accelerator pedal to increase the car's velocity.
- A car setting off from the traffic lights. There is an instant when the car is both stationary and accelerating. Otherwise it would not start moving.
- A car travelling round a bend at a steady speed. The car's speed is constant, but its velocity is changing as it changes direction.



(i)



(ii)

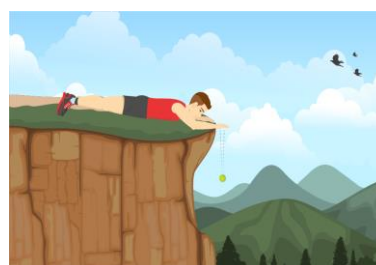


(iii)

- (iv) A ball being hit by a tennis racket. Both the ball's speed and direction are changing. The ball's velocity changes.
- (v) A stone dropped over a cliff. Gravity makes the stone go faster and faster. The stone accelerates as it falls.



(iv)



(v)

- If the velocity of an object increases with time, such a motion is called '**accelerated motion**'. In such motion, acceleration 'a' is considered positive for numerical problems.

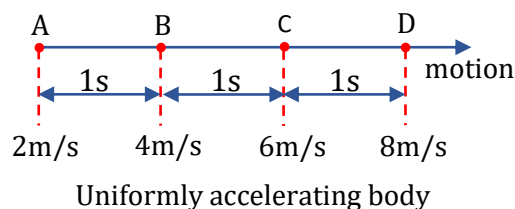
Example : An object starts from rest and its velocity goes on increasing with time.

- If velocity of an object decreases with time, such a motion is called '**retarded motion**'. In such a motion, acceleration is called '**retardation**' or '**deceleration**' and it is considered negative for numerical problems.

Example : A vehicle in motion is stopped by applying brakes.

Uniform acceleration

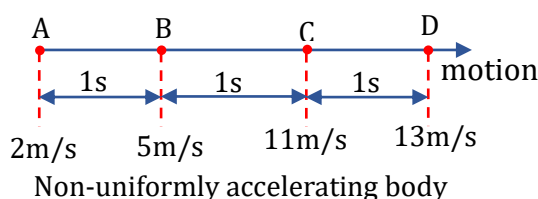
When a body moving in a straight line undergoes equal changes of velocity in equal intervals of time, the body is said to be moving with a uniform acceleration. Also, uniform acceleration means an acceleration having a constant magnitude and a constant direction (see figure).



- Examples :** (i) Motion of a freely falling body.
(ii) Motion of a ball rolling down on an inclined plane.

Non-uniform acceleration or variable acceleration

When a body undergoes unequal changes of velocity in equal intervals of time, the body is said to be moving with non-uniform acceleration (see figure).



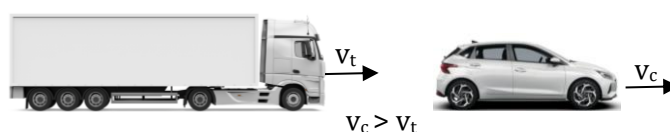
- Examples :** (i) The motion of a bus leaving or entering the bus stop.
(ii) A car moving on a busy road has non-uniform acceleration.



1. At a given instant of time, a car and a truck are travelling side by side in adjacent lanes of a highway. The car has a greater velocity than the truck. Does the car necessarily have a greater acceleration? Explain.

Explanation

The car may or may not have a greater acceleration. The instantaneous velocity of the car is greater than that of the truck, but its acceleration may be positive, negative or even zero. The acceleration of car may be less than that of truck (see figure).



2. Two cars are moving on straight sections of a highway. The acceleration of the first car A is greater than the acceleration of the second car B and both accelerations have the same direction. Which one of the following is true?
- The velocity of the first car is always greater than the velocity of the second car.
 - The velocity of the second car is always greater than the velocity of the first car.
 - In the same time interval, the velocity of the first car changes by a greater amount than the velocity of the second car does.
 - In the same time interval, the velocity of the second car changes by a greater amount than the velocity of the first car does.

Explanation

Option (c) is true because the acceleration of the first car is greater than the acceleration of the second car, thus in the same time interval, the velocity of the first car changes by a greater amount than the velocity of the second car does.

Option (d) is reverse of option (c), thus it cannot be true simultaneously. It is therefore, a false statement.

Option (a) is false because initial velocity of car A may be less than the velocity of car B. After a certain time interval, velocity of A will become more than the velocity of B.

Option (b) is also false because initial velocity of car A may be less or more than the velocity of car B. Even if the initial velocity of car B is more than velocity of car A, after a certain time interval it will become less than that of car A.



1. A scooter acquires a speed of 36 km/h in 10 s after the start. Calculate the acceleration of the scooter.

Solution

Given, Initial velocity, $u = 0$

$$\text{Final velocity, } v = 36 \text{ km/h} = \frac{36 \times 1 \text{ km}}{1 \text{ h}} = \frac{36 \times 5 \text{ m}}{18 \text{ s}} = 10 \text{ m/s}$$

Time taken, $t = 10 \text{ s}$

$$a = \frac{v - u}{t} = \frac{10 \text{ m/s} - 0 \text{ m/s}}{10 \text{ s}} = \frac{10 \text{ m/s}}{10 \text{ s}} = 1 \text{ m/s}^2$$

Acceleration = **1 m/s²**

2. The velocity of a runner is 10 m/s. After 4 seconds, his velocity becomes 20 m/s. What is the value of acceleration?

Solution

Given, initial velocity, $u = 10 \text{ m/s}$

final velocity, $v = 20 \text{ m/s}$; time, $t = 4 \text{ seconds}$

$$a = \frac{v - u}{t} = \frac{20 \text{ m/s} - 10 \text{ m/s}}{4 \text{ s}} = \frac{10 \text{ m/s}}{4 \text{ s}} = 2.5 \text{ m/s}^2$$

Acceleration = **2.5 m/s²**

When a body falls freely under gravity, the acceleration produced in the body due to Earth's gravitational attraction is called acceleration due to gravity.

Equations of uniformly accelerated motion

When an object moves with a uniform acceleration, its motion is called '**uniformly accelerated motion**'.

In case of uniformly accelerated motion, the average velocity is given by

$$V_{av} = \frac{v + u}{2}, \text{ here } v = \text{final velocity, } u = \text{initial velocity}$$

These equations give the relationship between initial velocity, final velocity, time taken, acceleration and distance travelled by the body.

First equation of motion

A body having an initial velocity 'u' acted upon by a uniform acceleration 'a' for time 't' such that final velocity of the body is 'v'.

$$\text{Acceleration} = \frac{\text{Change in velocity}}{\text{Time taken}} = \frac{\text{Final velocity} - \text{Initial velocity}}{\text{Time taken}}$$

$$a = \frac{v - u}{t}, \quad \text{or} \quad v - u = at, \quad \text{or} \quad \boxed{v = u + at}$$

Second equation of motion

It gives the distance travelled by a body in time 't'.

A body having an initial velocity 'u' acted upon by a uniform acceleration 'a' for time 't' such that final velocity of the body is 'v' and the distance covered is 's'.

$$V_{av} = \frac{v+u}{2}$$

Distance covered = average velocity $\times$ time taken

$$s = \left(\frac{v+u}{2} \right) \times t \quad \dots (1)$$

but $v = u + at$ (from first equation of motion) $\dots (2)$

Using (1) and (2), we get, $s = \left(\frac{u+at+u}{2} \right) t = \left(\frac{2u}{2} + \frac{at}{2} \right) t = \left(u + \frac{at}{2} \right) t$

or $\boxed{s = ut + \frac{1}{2}at^2}$

Third equation of motion

A body having an initial velocity 'u' moving with a uniform acceleration 'a' for time 't' such that final velocity is 'v' and the distance covered is 's'.

$$V_{av} = \frac{v+u}{2}$$

Distance covered = average velocity $\times$ time taken

$$s = \left(\frac{v+u}{2} \right) \times t \quad \dots (1)$$

Now, $v = u + at$, or $v - u = at$

or $t = \left(\frac{v-u}{a} \right) \quad \dots (2)$

From (1) and (2), we get,

$$s = \left(\frac{v+u}{2} \right) \left(\frac{v-u}{a} \right) = \frac{v^2 - u^2}{2a}$$

or $2as = v^2 - u^2$ or $\boxed{v^2 = u^2 + 2as}$



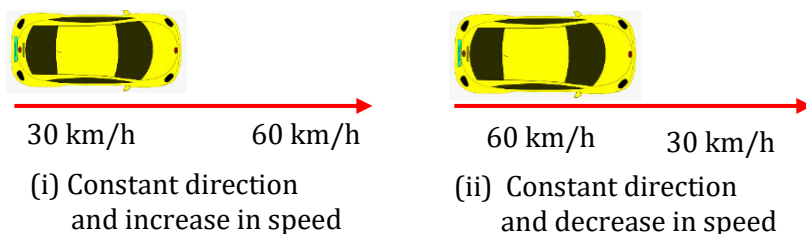
In your everyday life, you come across a range of motions in which

- (a) Acceleration is in the direction of motion.
- (b) Acceleration is against the direction of motion.
- (c) Acceleration is uniform.
- (d) Acceleration is non-uniform.

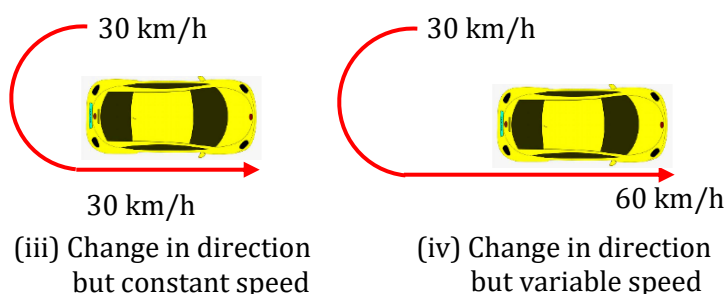
Identify one example each for the above types of motion.

Solution

- (a) While increasing the speed of vehicle using the accelerator, the acceleration is in the direction of motion. [see figure (i)]
- (b) While applying brakes of a vehicle, its speed decreases with time. Here, the acceleration is against the direction of motion. [see figure (ii)]



- (c) Motion of a particle under gravity has a uniform acceleration ($g = 9.8 \text{ m/s}^2$, vertically downwards).
- (d) Motion of a car in a crowded traffic has non-uniform acceleration as its speed varies (increases or decreases) as per the need. [see figure (iii) and (iv)]



- 1. A car accelerates uniformly from 18 km h^{-1} to 36 km h^{-1} in 5s. Calculate (i) acceleration, (ii) distance covered by the car in that time.**

Solution

Given, $u = 18 \text{ km/hr} = 5 \text{ m/s}$; $v = 36 \text{ km/h} = 10 \text{ m/sec}$; $t = 5 \text{ s}$

$$(i) \quad a = \frac{v - u}{t} = \frac{10 \text{ m/s} - 5 \text{ m/s}}{5 \text{ s}}$$

$$a = 1 \text{ m/s}^2$$

$$(ii) \quad s = ut + \frac{1}{2}at^2$$

$$s = 5 \times 5 + \frac{1}{2} \times 1 \times 5 \times 5 = s = 25 \text{ m} + 12.5 \text{ m} = 37.5 \text{ m}$$

2. A car initially at rest starts moving with a constant acceleration of 0.5 m s^{-2} and travels a distance of 25 m. Find its final velocity.

Solution

Given, $u = 0$; $a = 0.5 \text{ m/s}^2$; $s = 25 \text{ m}$

$$v^2 = u^2 + 2as$$

$$v^2 = (0)^2 + 2 \times 0.5 \times 25 = 25$$

$$v = \sqrt{25} = 5 \text{ m/s}$$

3. A racing car has a uniform acceleration of 5 m/s^2 . What distance will it cover in 20 s after starting from rest?

Solution

The car starts from the position of rest. So, its initial velocity (u) is zero. Then,

$u = 0 \text{ m/s}$; $a = 5 \text{ m/s}^2$; $t = 20 \text{ s}$, $s = ?$

we know that, $s = ut + \frac{1}{2}at^2$ (second equation of motion)

$$\text{or, } s = (0 \text{ m/s} \times 20 \text{ s}) + \frac{1}{2} \times (5 \text{ m/s}^2) \times (20 \text{ s})^2$$

$$\text{or, } s = 0 + \frac{1}{2} \times 5 \times 20 \times 20$$

$$\text{or, } s = 5 \times 10 \times 20 = 1000 \text{ m} = 1 \text{ km}$$

So, the car will cover a distance of 1000 m (or 1 km) in 20 s.

4. A body has a velocity of 3 m/s . If it has a uniform acceleration of 10 m/s^2 . Find the distance travelled by it in 3 seconds.

Solution

$u = 3 \text{ m/s}$; $a = 10 \text{ m/s}^2$; $t = 3 \text{ s}$, $s = ?$

we know that, $s = ut + \frac{1}{2}at^2$ (second equation of motion)

$$\text{or, } s = (3 \text{ m/s} \times 3 \text{ s}) + \frac{1}{2} \times (10 \text{ m/s}^2) \times (3 \text{ s})^2$$

$$\text{or, } s = 9 + (5 \times 9) = 9 + 45 = 54 \text{ m}$$

The distance travelled by body in 3 seconds is **54 m**.

5. If a body has a velocity of 50 m/s , and it takes 4 seconds to travel a distance of 240 m, then what is the acceleration of the body?

Solution

Given, initial velocity, $u = 50 \text{ m/s}$; time, $t = 4 \text{ s}$; distance, $s = 240 \text{ m}$; acceleration, $a = ?$,

we know that, $s = ut + \frac{1}{2}at^2$ (second equation of motion)

$$\text{or, } 240 = (50 \times 4) + \left(\frac{1}{2} \times a \times 16\right); 240 = 200 + 8a; 40 = 8a \text{ or, } a = 5 \text{ m/s}^2$$

6. The speed of a car is reduced from 54 km/h to 36 km/h in a certain time during which it travelled 125 m. Calculate the acceleration of the car.

Solution

$$\text{Given, initial velocity, } u = 54 \text{ km/h} = \frac{54 \times 1 \text{ km}}{1 \text{ h}} = \frac{54 \times 5 \text{ m}}{18 \text{ s}} = 15 \text{ m/s}$$

$$\text{final velocity, } v = 36 \text{ km/h} = \frac{36 \times 1 \text{ km}}{1 \text{ h}} = \frac{36 \times 5 \text{ m}}{18 \text{ s}} = 10 \text{ m/s}$$

Distance, $s = 125 \text{ m}$; acceleration, $a = ?$

$$v^2 = u^2 + 2as \quad (\text{from third equation of motion})$$

$$a = \frac{v^2 - u^2}{2s} = \frac{(10)^2 - (15)^2}{2 \times 125} = \frac{100 - 225}{250} = \frac{-125}{250} = -0.5$$

acceleration, $a = -0.5 \text{ m/s}^2$

7. A train starting from rest attains a speed of 90 km/hr. If the acceleration of the train is 0.5 m/s^2 , how much distance will it travel?

Solution

Given, initial velocity, $u = 0$; acceleration, $a = 0.5 \text{ m/s}^2$

$$\text{final velocity, } v = 90 \text{ km/h} = \frac{90 \times 1 \text{ km}}{1 \text{ h}} = \frac{90 \times 5 \text{ m}}{18 \text{ s}} = 25 \text{ m/s}$$

$$v^2 = u^2 + 2as \quad (\text{from third equation of motion})$$

$$(25 \text{ m/s})^2 = 0 + 2 \times 0.5 \times s$$

$$625 = 0 + s$$

$$s = 625 \text{ m}$$

8. The driver of a train travelling at 40 ms^{-1} applies the brakes as a train enters a station. The train slows down at a rate of 2 ms^{-2} . How long will the train travel before it stops?

Solution

Given, $u = 40 \text{ m/s}$, $v = 0$, $a = -2 \text{ m/s}^2$ (negative sign is taken as the velocity of object decreases with time and hence it is a retarded motion)

$$v^2 = u^2 + 2as \quad (\text{using 3rd equation of motion})$$

$$(0)^2 = (40 \text{ m/s})^2 + 2 \times (-2 \text{ m/s}^2) \times s$$

$$4 \text{ m/s}^2 \times s = 40 \times 40 \text{ m}^2/\text{s}^2$$

$$s = \frac{1600}{4}$$

$$s = 400 \text{ m}$$

Graphical representation of motion

Graph

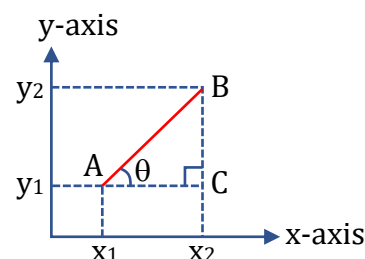
A graph is a line, straight or curved, showing the relation between two variable quantities, of which one varies as a result of the change in the other.

The quantity which changes independently is called **independent variable** and the one which changes as a result of the change in the other is called **dependent variable**.



$$\text{Slope} = \tan \theta = \frac{\text{Perpendicular}}{\text{base}} = \frac{\text{Change in y-axis}}{\text{Change in x-axis}}$$

$$\text{Slope of AB} = \frac{BC}{AC} = \frac{y_2 - y_1}{x_2 - x_1}$$

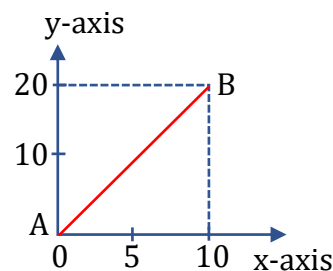


1. The graph shown below is straight line graph inclined to x-axis. What is the slope of the graph?

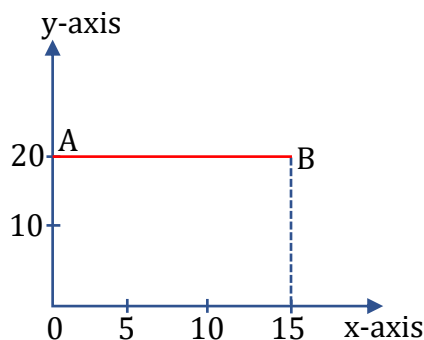
Solution

$$\text{Slope of line AB} = \frac{\text{Perpendicular}}{\text{base}} = \frac{\text{Change in y-axis}}{\text{Change in x-axis}}$$

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{20 - 0}{10 - 0} = \frac{20}{10} = 2$$



2. The graph shown below is straight line graph parallel to x-axis. What is the slope of the graph?

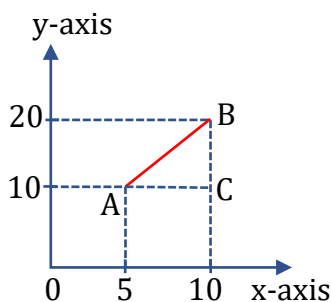


Solution

$$\text{Slope of line AB} = \frac{\text{Perpendicular}}{\text{base}} = \frac{\text{Change in y-axis}}{\text{Change in x-axis}}$$

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{20 - 20}{15 - 0} = \frac{0}{15} = 0$$

3. Find the slope of the graph in the given figure.



Solution

$$\text{Slope of line AB} = \frac{\text{Perpendicular}}{\text{base}} = \frac{\text{Change in y - axis}}{\text{Change in x - axis}}$$

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{20 - 10}{10 - 5} = \frac{10}{5} = 2$$

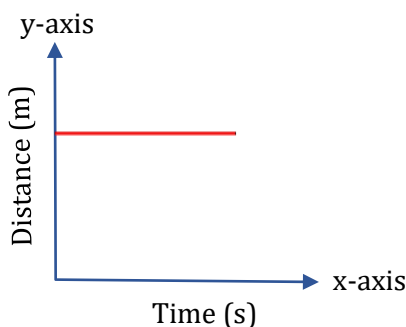
Distance/displacement-time graph

This graph is plotted between the time taken and the distance covered. The time is taken along the x-axis and the distance covered is taken along the y-axis.

- ▶ The slope of the distance-time graph gives the speed of the body.
- ▶ The slope of the displacement-time graph gives the velocity of the body.

When the body is at rest

When position of the body does not change with time then it is said to be stationary. The distance-time graph of such a body is a straight line parallel to x-axis (see figure).



s-t graph for a body at rest

When the body is in uniform motion

In uniform motion, $v = \text{constant}$.

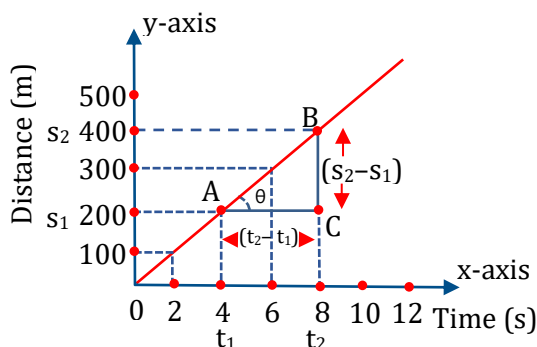
$$\text{Now, } s = v \times t$$

Since v is constant

$\therefore s \propto t$, thus, the distance-time graph of such a body is a straight line, inclined to x-axis.

$$\text{Slope} = \frac{\text{measure on y-axis}}{\text{measure on x-axis}} = \tan \theta \quad \text{or} \quad \text{Slope} = \frac{s_2 - s_1}{t_2 - t_1} = v$$

Thus, slope of distance-time graph gives speed of the body (see figure).

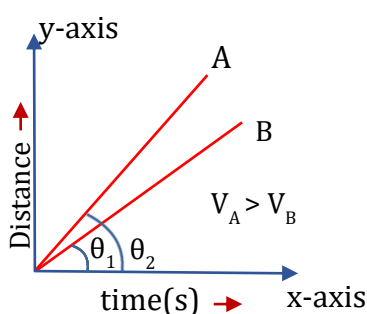


s-t graph for a body in uniform motion ($a = 0$)

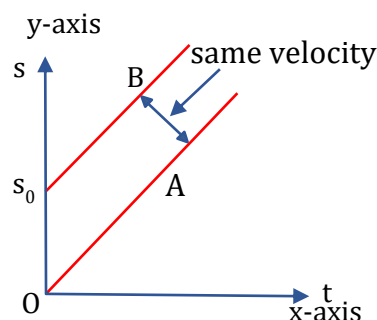
- In a distance-time graph, more the slope of the graph of an object in motion, more will be its speed and vice-versa.

Example : In given graph [see figure (a)], speed of particle A is greater than the speed of particle B because slope of graph of particle A is greater than slope of graph of particle B.

- In the given graph [see figure (b)], speed of particle A is equal to speed of particle B because graphs of both have same slope.



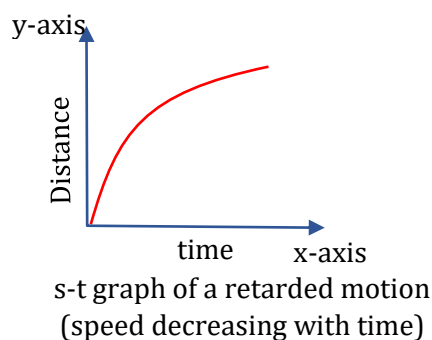
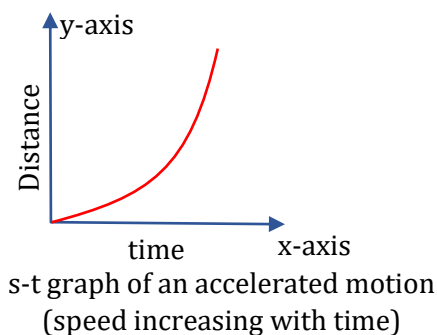
(a) Two particles moving with different velocities



(b) Two particles moving with same velocities

When the body is in non-uniform motion

In this case, distance-time graph is a curve. For example, in uniformly accelerated motion, $s = ut + \frac{1}{2}at^2$. Since $s \propto t^2$, definitely the graph is not a straight line, it is a curve (see figure).



- A distance-time graph can never be parallel to y-axis (representing distance) because this line has inclination of 90° , and slope = $\tan \theta = \tan 90^\circ = \text{infinite}$, which means infinite speed; it is impossible.

Speed/velocity-time graph

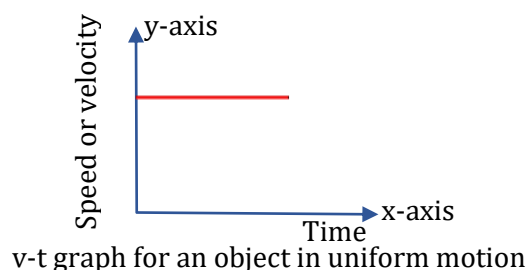
The variation in velocity with time for an object moving in a straight line can be represented by a velocity-time graph. In this graph, time is represented along the x-axis and velocity is represented along the y-axis.

The slope of the speed/velocity-time graph gives the acceleration of the body.

Area enclosed under a speed-time graph or velocity-time graph gives the distance covered by the body.

When the body is in uniform motion

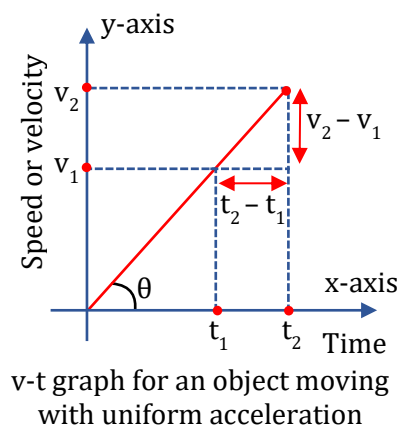
Here, the body moves with constant velocity. The speed or velocity of the body is uniform, hence the magnitude remains same. The graph is a straight line parallel to x-axis (time-axis). Since the velocity is uniform, its acceleration is zero. The slope of the graph in this case is zero (see figure).



When the body is moving with a uniform acceleration

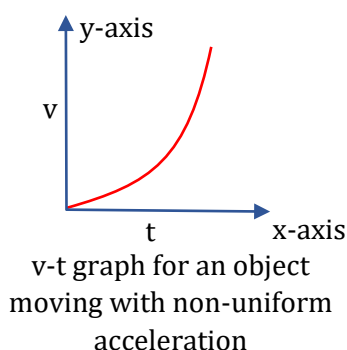
$$\text{Slope} = \frac{v_2 - v_1}{t_2 - t_1} = \tan \theta = a$$

Thus, slope of v-t graph gives acceleration of a body (see figure).



When the body is moving with a non-uniform (variable) acceleration.

In this case, the speed or velocity-time graph is not a straight line, but is a curve (see figure)



- Speed or velocity-time graph line can never be parallel to y-axis (speed axis), because inclination becomes 90° , then, $\text{slope} = \tan \theta = \tan 90^\circ = \text{infinite}$; i.e. infinite acceleration is impossible.

Distance from speed or velocity-time graph

As distance = speed $\times$ time, hence the distance can be calculated from speed or velocity-time graph.

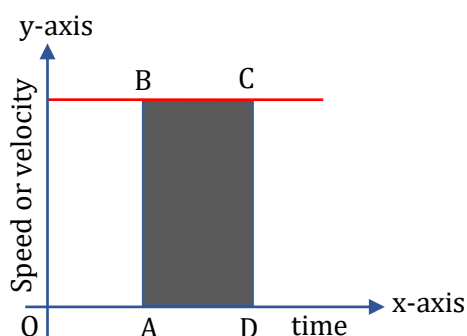
When speed or velocity is uniform (constant)

Distance = Area of rectangle ABCD = AB $\times$ AD [see figure (a)]

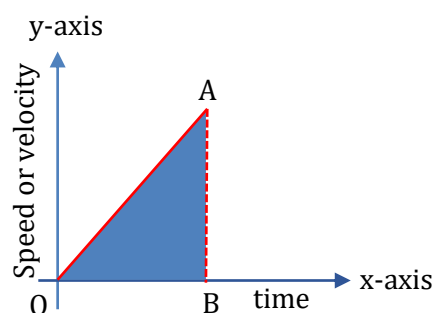
When acceleration is uniform (constant)

Distance or displacement = Area of right-angled triangle OAB [see figure (b)]

$$= \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times \text{OB} \times \text{BA}$$



(a)

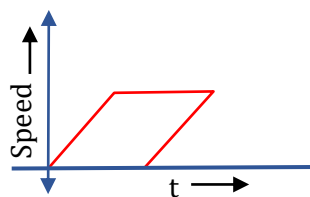


(b)

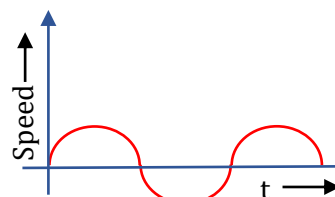
Area under v-t graph gives distance travelled by a body



1. State with reasons which of these cannot possibly represent one-dimensional motion of a particle (see figure).



(a)



(b)

Solution

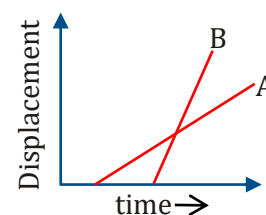
(1) Graph (a) is incorrect as time always increases, it cannot be reversed.

(2) Graph (b) is also incorrect as speed is always positive, it can never be negative.

2. Figure shows distance-time graph of two objects A and B. Which object is moving with greater speed when both are moving?

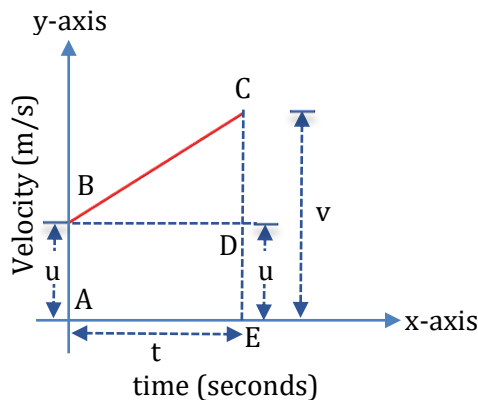
Solution

The line for object B makes a larger angle with the time-axis. Its slope is larger than the slope of the line for object A. Thus, the speed of B is greater than that of A.



Equations of motion by graphical method

Figure below represents a velocity-time graph, in which AB represents the initial velocity u , CE represents final velocity v , such that the change in velocity is represented by CD, which takes place in time t , represented by AE (see figure).



An object moving with certain initial velocity u and attains final velocity v after time t .

Derivation of $v = u + at$

Acceleration = slope of the graph line BC

$$a = \frac{CD}{BD} = \frac{CE - DE}{BD} \quad \text{or} \quad a = \frac{v - u}{t}$$

$$\left[\begin{array}{l} \because CE = v \\ \because DE = AB = u \\ \because BD = AE = t \end{array} \right]$$

$$v - u = at \quad \text{or} \quad \boxed{v = u + at}$$

Derivation of $s = ut + \frac{1}{2}at^2$

Distance travelled = Area of trapezium ABCE

= Area of rectangle ABDE + Area of triangle BCD

$$= AB \times AE + \frac{1}{2} (BD \times CD) = u \times t + \frac{1}{2} [t \times (v - u)]$$

$$= u \times t + \frac{1}{2} [t \times (u + at - u)]$$

$$\left[\begin{array}{l} \because BD = AE = t \\ \because CD = CE - DE = v - u \end{array} \right]$$

$$[\because v = u + at]$$

$$\boxed{s = ut + \frac{1}{2}at^2}$$

Derivation of $v^2 = u^2 + 2as$

From the velocity-time graph, distance covered = Area of trapezium ABCE

$$\Rightarrow s = \frac{1}{2} (AB + CE) \times AE \quad \therefore s = \frac{1}{2} (u + v) \times t \quad \dots(i)$$

$$\text{Acceleration} = \frac{\text{Change in velocity}}{\text{Time}}$$

$$a = \frac{v - u}{t} \quad \therefore t = \frac{v - u}{a} \quad \dots(ii)$$

Substituting the value of t in equation (i)

$$s = \frac{(v + u)}{2} \times \frac{(v - u)}{a} \quad [\because A^2 - B^2 = (A + B) \times (A - B)]$$

$$s = \frac{v^2 - u^2}{2a}, \quad \text{or} \quad v^2 - u^2 = 2as, \quad \text{or} \quad \boxed{v^2 = u^2 + 2as}$$

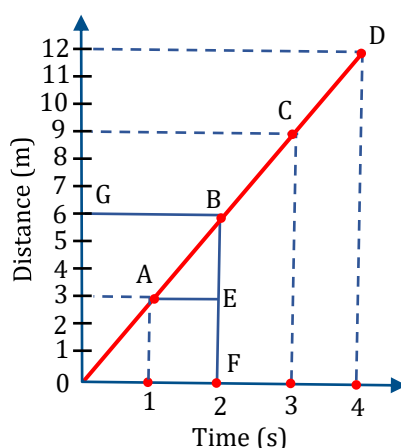


1. The distance covered by a moving body was recorded after every 1 s for 4 s. The data is given in the table below. Represent it graphically and find the speed of the body from the graph.

Distance (m)	0	3	6	9	12
Time (s)	0	1	2	3	4

Solution

The body covers equal distance in equal interval of time. The distance-time graph of a body in uniform motion is a straight line. The horizontal axis represents time, while the vertical axis represents distance (see figure).

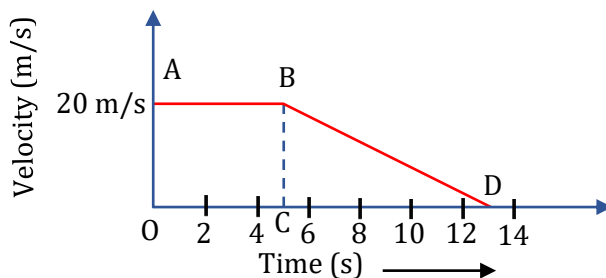


$$\text{Speed} = \text{Slope} = \frac{\text{measure on y-axis}}{\text{measure on x-axis}} = \frac{OG}{OF} = \frac{6\text{m}}{2\text{s}} = 3 \text{ m/s}$$

If we study the graph carefully, we will see that BE/AE would also give the speed of the body. BE gives the distance travelled in the time interval AE , which is 1 s.

2. A car travels with a uniform velocity of 20 ms^{-1} for 5s. The brakes are then applied and the car is uniformly retarded. It comes to rest in further 8 s. Draw a graph of velocity against time. Use this graph to find :

- (i) the distance travelled in first 5 s,
(ii) the distance travelled after the brakes are applied,



Solution

- (i) The distance travelled in first 5 seconds = area of rectangle OABC
area of rectangle = length $\times$ breadth = $20 \text{ m/s} \times 5 \text{ s} = \mathbf{100 \text{ m}}$
(ii) The distance travelled in last 8 seconds (from 5 sec to 13 sec) = area of the triangle

$$BCD = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times (13 - 5) \text{ s} \times 20 \text{ m/s}$$

$$\text{or } \frac{1}{2} \times 8 \text{ s} \times 20 \text{ m/s} = \mathbf{80 \text{ m}}$$

3. In the above question, find the acceleration during the first 5 s and last 8 s.

Solution

- (i) Acceleration during first 5 second is zero as velocity is constant.

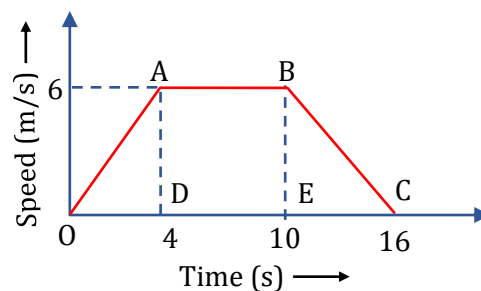
$$a = \frac{v - u}{t} = \frac{20 - 20}{5} = \mathbf{0 \text{ m/s}^2}$$

- (ii) In the last 8 seconds, acceleration,

$$a = \frac{v - u}{t} = \frac{0 - 20}{13 - 5} = \frac{20}{8} = \mathbf{-2.5 \text{ m/s}^2}$$

4. Study the speed-time graph of a body given here and answer the following questions:

- (a) What type of motion is represented by OA?
(b) What type of motion is represented by AB?
(c) What type of motion is represented by BC?



Solution

- (a) OA is a straight line graph between speed and time. The slope of speed-time graph represent acceleration. Thus, OA represent uniformly accelerated motion.

- (b) AB is a straight line parallel to the time axis. The speed is constant from A to B. Thus, AB represent uniform speed.
- (c) BC is a straight-line graph between speed and time (downward slope). BC represents uniform retardation or negative acceleration (uniformly retarded motion).

5. In the above question, find the acceleration, retardation and the distance travelled by the body.

Solution

- (i) In the graph shown, speed of the body increases from 0 to 6 m/s in the first 4 seconds.

Acceleration = slope of line OA

$$a = \frac{v_2 - v_1}{t_2 - t_1} = \frac{6 - 0}{4 - 0} = 1.5 \text{ m/s}^2$$

- (ii) In the graph shown, speed of the body decreases from 6 m/s to 0 in the last 6 seconds.

Retardation = slope of line BC

$$a = \frac{v_2 - v_1}{t_2 - t_1} = \frac{0 - 6}{16 - 10} = -1 \text{ m/s}^2$$

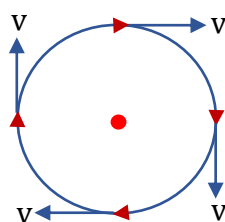
- (iii) Distance travelled by the body = area of shaded portion

= area of triangle OAD + area of rectangle DABE + area of triangle BEC

$$= \left(\frac{1}{2} \times 4 \times 6 \right) + (6 \times 6) + \left(\frac{1}{2} \times 6 \times 6 \right) = 66 \text{ m}$$

Circular motion

When a particle moves along a circular path, its motion is called '**circular motion**' (see figure).



Circular motion of an object

- In a circular motion, velocity of a particle is tangential to the circular path.

If the body covers equal distances along the circumference of the circle, in equal intervals of time, then motion is said to be a **uniform circular motion**. When a body moves along a circular path, then its direction of motion changes continuously. Thus, **a circular motion is always a non-uniform motion**.

- A uniform circular motion is a motion in which speed remains constant but direction of velocity changes continuously.

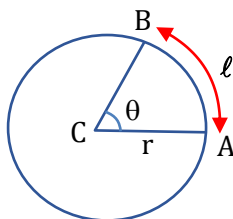
Examples of uniform circular motion

- (i) An athlete running on a circular track with constant speed.
- (ii) Motion of tips of the second hand, minute hand and hour hand of a wrist watch.

Radian

It is the SI unit for measuring plane angles in physics.

The arc AB of the circle has length ℓ and subtends an angle θ at the centre C (see figure).



Plane angle in terms of radian

If $\angle ACB = \theta$ radians

Then, $\theta = \frac{\text{arc}}{\text{radius}}$

$$\theta = \frac{\ell}{r}$$

when $\ell = r$, then $\theta = 1$ radian.

One radian is defined as the angle subtended at the centre of the circle by an arc which is equal in length to its radius.

Angle subtended by the circumference at the centre,

$$\theta = \frac{2\pi r}{r} = 2\pi \text{ radians}; 2\pi \text{ radians} = 360^\circ; 1 \text{ radian} = \frac{360^\circ}{2\pi}$$

$$1 \text{ radian} = 57.3^\circ$$

Angular displacement and angular velocity

The angle covered by a body moving along a circular path in a given interval of time is called '**angular displacement**'. It is denoted by θ .

The angle covered by a body per unit time is called '**angular velocity**'. It is usually denoted by ω and measured in radians per sec. If θ is the angle covered in time 't' then,

$$\text{Angular velocity} = \frac{\text{Angular displacement}}{\text{Time taken}}$$

$$\therefore \omega = \frac{\theta}{t}$$

Unit

Angular displacement θ : radian; Angular velocity ω : rad/s

Relation between linear speed and angular velocity

Let us consider a particle moving along a circular path of radius ' r ' with constant angular velocity ' ω '. Let it covers an angle ' θ ' during a time interval ' t '. Let the distance covered during time interval ' t ' be ' ℓ ' (see figure)

$$\text{Angular displacement} = \theta = \frac{\ell}{r}$$

$$\ell = r\theta \quad \dots (1)$$

$$\text{Also, angular velocity, } \omega = \frac{\theta}{t} \quad \dots (2)$$

$$\text{Now, linear speed, } v = \frac{\ell}{t}$$

$$\text{or } v = \frac{r\theta}{t} \quad [\text{using (1)}]$$

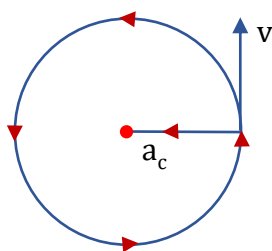
$$\text{or } v = r \left(\frac{\theta}{t} \right)$$

$$\text{or } \boxed{v = r\omega} \quad [\text{using (2)}]$$

Also, linear speed = $\frac{2\pi r}{T}$, where ' T ' is the time period for 1 revolution.

Centripetal acceleration

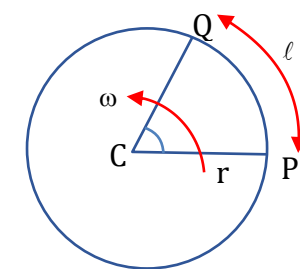
Circular motion is always accelerated, and it has a radially inward acceleration called 'centripetal acceleration'.



Circular motion always has radially inward acceleration

$$\boxed{a_c = \frac{v^2}{r}}$$

$$\text{Also, } a_c = \frac{(r\omega)^2}{r} = \frac{r^2\omega^2}{r} = r\omega^2$$



An object in circular motion